Multimedia Tools and Applications (2021) 80:753–771
https://doi.org/10.1007/s11042-020-09567-1

Tea leaf disease detection using multi-objective image segmentation

Somnath Mukhopadhyay[1] · Munti Paul[1] · Ramen Pal[1] · Debashis De[2]

Received: 10 December 2019 / Revised: 28 July 2020 / Accepted: 6 August 2020 /
Published online: 4 September 2020

Abstract

Tea leaves' diseases caused by constant exposure to pathogens lead to significant crop yield loss globally. Diagnosing the tea leave disease at an early stage minimizes the tea yield loss. In this study, a novel approach is presented for automatically detecting tea leaves diseases based on image processing technology. The Non-dominated Sorting Genetic Algorithm (NSGA-II) based image clustering is proposed for detecting the disease area in tea leaves. After that, PCA and multi-class SVM is used for feature reduction and identifying the disease in the tea leaves, respectively. The result shows that the proposed algorithm can detect the type of disease persisting in tea leaves with an average accuracy of 83%. Five different tea leaf diseases are considered here, such as Red Rust, Red Spider, Thrips, Helopeltis, and Sunlight Scorching.

Keywords Tea leaf disease detection · NSGA-II based image clustering · Multi-class SVM · PCA · Feature reduction

1 Introduction

Identification of diseases in plants is essential to avert the losses in yield and quantity in agriculture. Health monitoring and disease identification of plants is necessary for a sustainable cultivation system. The study of plant diseases means the knowledge of visible patterns

✉ Somnath Mukhopadhyay
som.cse@live.com

Munti Paul
muntip7@gmail.com

Ramen Pal
ramen.pal673@gmail.com

Debashis De
dr.debashis.de@ieee.org

[1] Dept. of Computer Science, Engineering, Assam University Silchar, Silchar, India

[2] Dept. of Computer Science, Engineering, West Bengal University of Technology, Kolkata, India

seen on the crops and vegetation. In the early days, plant diseases were controlled and analyzed by experts in the fields. Introducing technological advancements into the agriculture system can help the farmers identify the conditions well in advance. As a result, the farmers will be able to rectify the yield loss.

In India, tea production plays a vital role, and it is an integral part of the Indian economy also. However, several diseases affect the proper growth of tea leaves; as a result, tea production is highly compromised. The problem can be solved if appropriate treatments or trimming are applied at an early stage on the contaminated leaf to stop the further increase of that. It is essential to develop an automated and technological system for detecting diseases as a preventative measure to help the farmers detect the tea leaf disease at the initial stage. In most cases, disease symptoms are visible on the leaves, stems, and fruits. The first symptom that determines that the leaf is diseased is by color changes from a green hue to other colors. A healthy leaf has a distinct color, whereas an unhealthy leaf has drastically different color compared to its original color. Many researchers have reported many types of research in the past for detecting leaf disease using digital image processing and classification techniques. Their approaches were similar to some extent. First, healthy and unhealthy leaf images are acquired using cameras. Then, disease areas that are the regions-of-interest (ROI) in this case are segmented or extracted from the background. Then different image textures like the color, shape, or other features are extracted. Finally, classification and clustering techniques are used to classify diseased images [1].

1.1 Related works

In this subsection, we have studied and reviewed different research contributions, models, opinions, thoughts, and ideas on leaf disease identifications. There are many kinds of research in the literature on technological developments in recent times in this problem area, including advances in technological resources, remote sensing technology, etc. In 2003, Helly et al. [14] proposed a method for Leaf disease detection where they used Hue Saturation Intensity (HSI) transformation followed by image segmentation using Fuzzy C-means. They used features like color, size, and shape of the unhealthy leaf portions, and finally, neural network-based classification is performed. Kim et al. [7] proposed a classification technique based grape-fruit peel disease detection method using the analysis of color and fine details. The features are extracted using spatial gray-level dependency matrices, and then classification using the squared distance method. Qing Yao et al. [61] proposed a rice disease detection algorithm using standard image processing techniques, followed by classification using support vector machine. They identified the diseased area and the corresponding shapes and texture features of the rice image. They showed that this method could effectively detect and classify the disease spots to an accuracy of more than 97%. T. Rumpf et al. [47] proposed a sugar beet leaves disease detection technique based on Support Vector Machine and spectral vegetation indices. Their objective was to differentiate unhealthy regions from healthy leaves, determine the Cercospora leaf region, leaf rust and powdery mildew, and detect diseases before any particular symptoms were visible. They used nine spectral vegetation indices, which are related to different features that are used for automatic classification. The classifications between healthy leaves and diseased leaves got accuracy more than 86%. Al-Bashish et al. [10] developed a method in which the leaf diseases are detected and classified using k-means based clustering and neural network-based classification. Phadikar et al. [46] proposed a classification-based rice-leaf disease detection technique that focuses on the basics of morphological changes in the image, such as leaf damage, covering blight and crop blast. Image clustering is done to detect and separate the diseased region on the image, and finally, Support Vector Machine based classification to detect the diseases. S. Arivazhagan et al. [2] developed a texture features-based technique to detect and classify the leaves' diseased spot. This approach has four steps: RGB image color transformation, masking and thresholding of green pixels, segmentation, and feature extraction, and finally, SVM-based classification. Omrani et al. [42] proposed a disease detection technique for the apple tree. The methodology contains a Support Vector Regression (SVR) technique, which uses radial basis functions as the kernel function. At first, the input leaf images have been transformed into device-independent color space (CIELAB) from the RGB image. After that, K-Means based image segmentation is performed to identify the diseased portion from the image. They also used techniques such as wavelet transformation and grey-level co-occurrence matrix conversion for such purpose. Features like color, shape, and texture of the leaf's unhealthy region are considered for training and classification using SVR. Sachin D. Khirade et al. [24] performed color transformation, then segmentation using k-means and finally, classification using ANN. B.C. Karmokar et al. [22] proposed a tea leaf disease detection technique. Starting with some image preprocessing such as cropping, resizing, and thresholding, then they performed a neural network-based ensemble technique for pattern recognition and classification. K. J. Mohan et al. [36] proposed an image processing system to detect the various paddy leaf diseases. They used features like Haar and AdaBoost classifier [35] to identify invalid portions of the paddy plant, which obtains an accuracy of 83.33%. Another method where the disease symptoms are detected using Scale Invariant Feature Transformation [28] and the classification is done using K-Nearest Neighbor [23] and Support Vector Machine (SVM). They achieved accuracies like 91% using SVM and 93% using K-NN. Pranjali B. Padol et al. [43] proposed a disease detection method on grape leaves using the SVM classifier. It starts with K-means-based clustering [40] followed by feature extraction and classification using SVM, which achieves an accuracy of more than 85%. Recently, Md. Selim Hossain et al. [18] reported a tea leaf disease detection using SVM with a reduced number of features. They achieved accuracy up to 93% as compared to other techniques. More recently, X Sun et al. [56] proposed a convolution neural network (CNN) based tea leaf disease detection technique. Initially, they performed image segmentation, and then the images are given to the network for training. They achieved better accuracy using CNN compared to SVM and BP neural network. In [9], Amar et al. proposed an image processing technique for betel vine rotten leaf where they used low size samples, and as a result, they obtained very low recall values. [54] considered plant leaves having matured symptoms for detecting the diseases. Shanwen et al. reported an article for leaf image-based cucumber disease recognition using sparse representation classification [62], which uses the K-Means algorithm for image segmentation. The algorithm obtained an accuracy of 85% but with high time complexity. Kamlesh et al. [17] reviewed the neural network-based disease detection techniques on hyperspectral image data. Jing et al. [5] proposed a tea leaf disease detection technique using a convolution neural network model. This algorithm is tested on large-sized data samples; however, the accuracy is above 80% on average. The algorithm is also computationally expensive. One similar algorithm proposed by Xiaoxiao et al. [55], which detects tea leaf diseases based on the convolution neural network model. This algorithm also suffers from high computational overhead. Some other recent attempts towards disease identification in leaves have been studied in [53, 63]. These algorithms are mainly based on image retrieval systems and some soft computing methodologies. In recent times, some image and video segmentation techniques are reported in the literature, such as graph-based High-order Energy Optimization (HEO) technique [52], which can be applied for segmenting images and motions in videos. HEO based graph-cut techniques try to minimize the cut and maximize the flow. Frameworks like Laplacian graph-based optimization [50], Generalized graph-cut technique based optimization [45], Quadratic Pseudo-Boolean Optimization(QPBO) [51] showed the effectiveness of HEO approach to bi-clusters the data set. A dynamic programming based framework to solve the problem mentioned above is also available in ([49]). The effectiveness of this algorithm depends on the selection of initial cluster centers. In recent years, Deep Neural Network(DNN) based approach is also showing it's effectiveness in solving image vision-related problems like image cropping [58], moving object tracking system [12, 27], video object segmentation [31, 32, 58] and image object co-segmentation. [58] proposed a clustering algorithm for super-pixel image segmentation using the DBSCAN algorithm; however, it requires two important input parameters, which are difficult to determine. The performance of the clustering output heavily depends on these user-supplied parameters. DBSCAN is used to separate clusters with high density with clusters with low density. However, we know from the leaf images that these are having varying densities. Separating high-density regions from low-density regions in tea leaves is difficult since density varies, in this case, very smooth.

1.2 Motivations and contributions

From the above studies, we can observe that many types of research are carried out in detecting the crop and leaf diseases in the recent past. The researchers followed methodologies such as preprocessing the images, which followed by segmenting the disease portions and finally classification using some training models for detecting and classifying the leaf diseases. However, we have seen the scope of research in tea leaf disease detection, which can effectively diagnose leaf diseases. It is also required that the algorithm runs with low execution overhead and can successfully identify the diseases with early-stage symptoms. The primary objective here is to develop a tool that can directly help the farmers, and so, here, leaf images are captured using a mobile camera. The second objective of this model is to identify the diseased portions in the tea leaf so that the disease diagnosis becomes easy. Another objective of this algorithm is to deal with a higher number of diseases from different tea gardens and to identify the disease names.

We have not performed too many operations at the preprocessing stage to reduce computational overhead except conversion from RGB to Hue Saturation (HS). The intensity band from the HSI color model is eliminated. The diseased area in the leaf images is spotted using a multi-objective genetic algorithm based image segmentation algorithm. The optimal set of features is identified by using Principle Component Analysis (PCA). Finally, a multi-class SVM is used for detecting the disease from the infected leaf. The proposed algorithm can identify five numbers of diseases from the input images and also suggests corresponding actions to be taken so that farmers can get complete help from this. In the results section, extensive experiments are performed to validate the proposed methodology. The silhouette index is used to validate the proposed clustering algorithm using NSGA-II. The results show that the multi-class SVM can effectively identify the diseases to a satisfactory accuracy.

1.3 Preliminaries

In this subsection, we brief basic ideas of three components of this research work, such as Principal Component Analysis, Support Vector Machine, and K-Means algorithm.

1.3.1 Principal component snalysis(PCA)

Principal Component Analysis (PCA) is an exploratory data investigation framework. It is applied to highlight the variation in the dataset and extract the dominant patterns from it [30]. It is used for dimensions reduction of the dataset. The dimension of the output data is a user-defined choice. In this approach, first, the data set is represented as a standardized matrix. Then the co-variance of that matrix is calculated. After that, the Eigenvalues and their corresponding Eigenvectors of that matrix are computed. Later by using the sorted array of Eigenvectors, the new features are identified [59].

1.3.2 Support vector machine(SVM)

Support Vector Machine is a supervised framework in the machine learning paradigm. This technique can efficiently solve any classification (binary or multi-class) and regression problem [13]. It usually plots the observations from a population into an n-dimensional space. Here n defines the number of features in the dataset. Then classification is performed by finding an optimal hyper-plane on that n-dimensional space to classify all the observations into two classes. The optimal hyperplane is selected by considering the margin value. A margin value is a distance between the nearest data point and the hyper-plane. A hyper-plane that has a higher margin value and can classify two distinct classes is selected [4] as the separating hyper-plane.

1.3.3 K-Means clustering

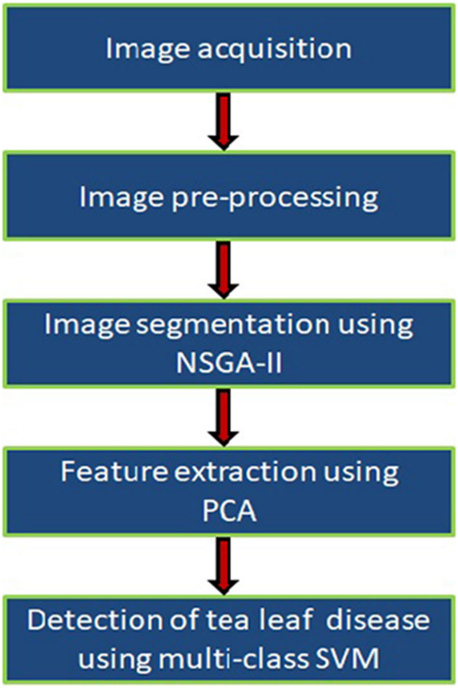
K-Means is a partitive clustering technique. It partitions a data set of size N into K distinct clusters. The value of K should be predefined [21]. Initially, a K number of observations from the data set is selected as centroids. The rest of the data set observations are then assigned to different clusters based on the squared distance between the observation and the centroid. The updated centroid of a cluster is computed using the arithmetic mean of all the observations that are members of that cluster. This process is iterated until there is no change in the centroids [33].

Later in Section 2, the proposed approach is described. In Section 3, results have been generated to validate the proposed algorithm. In Table 2, the tick/cross-comparison of the proposed model with fifteen other existing models is shown. In Fig. 8, the proposed algorithm is compared with K-Means based disease detection algorithm. Finally the conclusion is written in Section 4.

2 Proposed approach

The proposed algorithm has three steps. In the first step, captured the images of unhealthy tea leaves using a mobile camera and also preprocessing is performed. The disease area in the tea leave images is identified using NSGA-II based image clustering in the second step. In the third step, we have performed feature extraction using Principal Component Analysis, and a multi-class SVM algorithm is presented to identify the disease in the input leaf image. The detailed flow diagram of the proposed technique is depicted in Fig. 1. The proposed algorithm is clearly discussed in the following sections and subsections.

Fig. 1 Flow-diagram of the proposed methodology



2.1 Image acquisition and preprocessing

A mobile camera captures the plant leaves from three tea gardens of the Cachar district of Assam, viz., Silcoorie, Haticherra, and Rose Candy. Large numbers of image samples are collected, most of which are infected by Red Rust, Red Spider, Thrips, Helopeltis, and Sunlight Scorching diseases, and others are unaffected or otherwise healthy. Experts label the leaf images. The representation of these diseased leaves is shown in Fig. 2. It can be seen from these images that these five diseased images are having very sharp differences in their color and textures.

From the literature, it is known that image preprocessing is the most important operation [11], which is required to perform to get an image having suppressed unwanted distortions and also to highlight the image features which will be useful for subsequent processing. Much image preprocessing is avoided here to reduce the computational cost and use the

(a) Red Rust (b) RedSpider (c) Thrips (d) Helopeltis (e) Sun-lightScorching

Fig. 2 Various diseased tea leaf images of the study area application software on-the-go in the fields or gardens. Only an image-level conversion from the RGB model to Hue, Saturation, and Intensity (HSI) model is performed. After that, the intensity band is eliminated from the HSI image since the intensity band does not add any information for this application, instead only the hue color and textures are relevant here. The HS band images are shown in the results section.

2.2 Image segmentation

Here image segmentation is done to segment the different regions of tea leaf. Image segmentation [6, 20, 44] is the process of labeling each and every pixel in the image so that pixels with the same label share the same characteristics. In the following subsection, the multi-objective genetic algorithm based NSGA-II is performed for image clustering. The clustering output from the NSGA-II is used for image segmentation.

Among the available multi-objective optimization techniques, the genetic algorithm [15] based methods such as Non-dominated sorting GA-II (NSGA-II) [8], Strength Pareto evolutionary algorithm (SPEA), SPEA2 [64], and Pareto archived evolutionary strategy [25] are most popular. NSGA-II is an improved version over NSGA in terms of computational complexity. From [8], it can be seen that NSGA-II outperforms several other optimization algorithms. Hence, the multi-objective clustering technique is proposed here, which is based on NSGA-II.

2.2.1 NSGA-II based multi-objective clustering

The non-dominated sorting genetic algorithm is a multi-objective optimization technique from the field of evolutionary techniques [8, 26]. NSGA-II has the benefits of having less computational cost compared to NSGA, and it has also got the elitist model by merging the old and new generation chromosomes. In each generation, it propagates the non-dominated front($Front_1$) solutions to the next generation. This guarantees convergence and diversities of this algorithm. The crowded binary tournament selection algorithm is followed here as the selection algorithm. The complete flow diagram of the proposed NSGA-II based image clustering algorithm is given in Fig. 3. The algorithm starts by encoding a n number of initial chromosomes(P_i) in real numbers, with the coordinates of the random cluster centers. Each chromosome is encoded as $K \times d$, where K is the number of centers in the d dimension of the data points. The probability that a chromosome will be selected for reproduction is determined by its fitness value [41, 60]. In this NSGA-II based clustering algorithm, two objective functions are optimized, such as Intra-cluster and Inter-cluster distance functions. The *Euclidean* distance function($d(x)$) is used to compute the squared distance between two pixels [3]. The $d(x)$ [37, 38] between the i-th pixel(x_i) and j-th pixel(x_j) in the image is computed using (1).

$$d(x) = \sqrt{\Sigma_{i,j}(x_i - x_j)^2} \tag{1}$$

The intra-cluster distances of all the clusters are computed, and then the largest one out of that is taken in d_{max}, which is defined in (2). Here z is the partition matrix. z represents the belongingness of data points to the clusters(C_i). Here, N_c represents the total number of clusters. m_i represents the centroid values for the respective clusters and $|C_i|$ represents the cardinality of C_i. The Euclidean distance(d) between an observation(z_p) and it's corresponding cluster centroid(m_i) is calculated by using (1). In this bi-objective optimization algorithm, the intra-cluster distance is minimized, and the inter-cluster distance is maximized.

$$d_{max} = \max_{j=1 \ to \ N_c} \{ \sum_{\forall z_p \in C_{ij}} d(z_p, m_{ij}) / \ |C_{ij}| \} \tag{2}$$

Inter-cluster distances for all clusters are computed and the minimum one(d_{min}) is selected using (3). Here, the euclidean distance(d) between two cluster centroids(m_{ij_1} and m_{ij_2}) is calculated by using (1).

$$d_{min} = \min_{\forall j1, j2, j1 \neq j2} \{ d(m_{ij_1}, m_{ij_2}) \} \tag{3}$$

After that, non-dominated sorting is performed on the P_i to get the non-dominated fronts. A selection algorithm is then applied on the sorted P_i to create the mating pool of size n for reproduction. In the first iteration, Binary Tournament Selection (BTS) and in the rest, a crowded BTS are used as the selection methods. The ranks of individuals are considered in the binary tournament selection method. Whereas in the crowded binary tournament selection method, we considered ranks of individuals as well as their crowding distance values. Then reproduction such as crossover and mutation is performed. Here, general crossover and mutation operations are performed to reproduce the set of offsprings (Q_i). The size of Q_i is also n. Then P_i and Q_i are combined to generate a set(R_i) of combined offspring. The size R_i becomes $2n$. After that, non-dominated sorting is again performed on the R_i. After that the crowded binary tournament selection is performed on the sorted R_i to construct the next generation child chromosomes (P_{i+1}) of size n. The algorithm is executed for a constant number of iterations G with a population size of P. Finally, it produces a set of non-dominated solutions ($Front_1$). All the solutions this front are eligible to be selected as the set of optimal centroids. The solution(Sol_n), which has the lowest crowding distance value in ($Front_1$), is selected as the optimal set of cluster centers. Then segmentation is performed on the input image by using the Sol_n.

2.3 Disease detection and classification

The leaf disease detection and classification algorithms are divided into two parts, viz., feature selection and classification. From the segmented image, the required features are identified using Principle Component Analysis (PCA). Those features are then used to train a Support Vector Machine (SVM) [1] for the detection of diseases in the tea leaves.

2.3.1 Feature reduction and training

Feature reduction is one of the most vital phases of this algorithm. Feature reduction is used to select the optimal feature vector based on statistical tests [57]. Here, the features are selected very carefully based on their unique characteristics for the five types of diseases of tea leaves. Initially, a large number of features are taken from the segmented images, but eventually, few are selected by Principal Component Analysis [19] method. In this work, the following features are finally obtained and used for training, namely:

- Number of Cluster Centers (NoC)
- Inter-Cluster Distance
- Variance
- Entropy
- Skewness

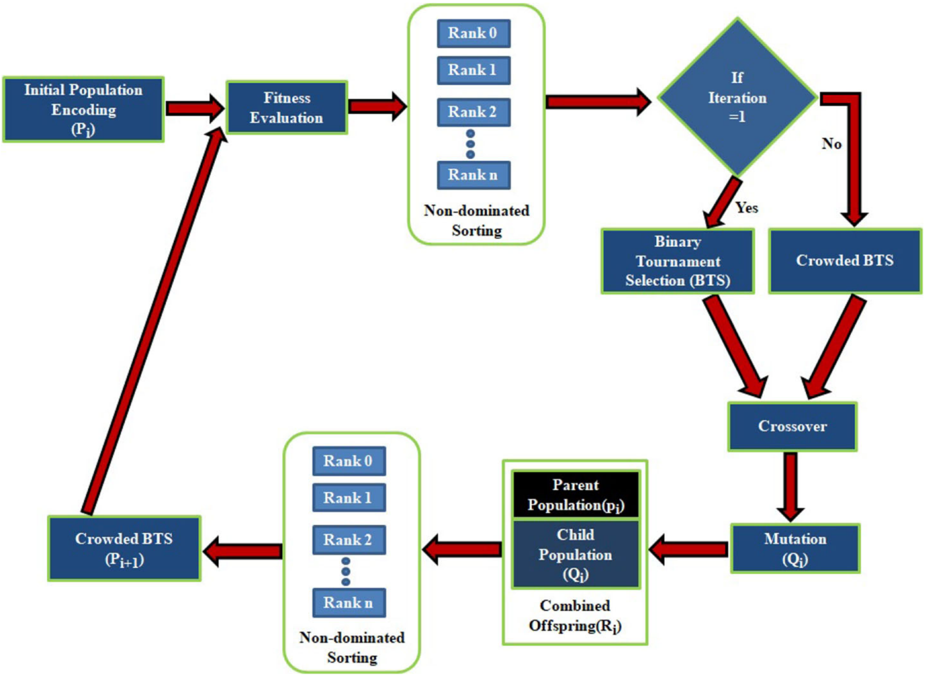


Fig. 3 Flow diagram of the NSGA-II based image clustering algorithm

Once the feature vector is finalized, training is performed using Support Vector Machine (SVM) for detecting the diseases in tea leaves. Keeping in mind the input data as being diseased or not in a d-dimensional space, an SVM based classification algorithm [39, 57] is considered. It computes a maximum-margin hyper plane which separates the binary classes of points in the space in either side of that and also on both sides of the hyper plane, two more hyper planes are computed, which are defined so that the distance between the hyper plane and the nearest points to either group is maximized. Originally, SVM is a two-class problem classifier. Here five number of two-class SVMs [34] are used because the problem here is to classify five different diseases. The following equation describes the SVM process:

$$\frac{1}{2}w^T w + C\sum_{1}^{N}\xi_i \tag{4}$$

Subject to the constraints:

$$y_i(w^T\phi(x_i) + b) \geq 1 - \xi,\ and\ \xi \geq 0, i = 1\ldots n \tag{5}$$

Let the data set has n features as $< x_i, y_i >$, where $y_i \in \{+1, -1\}$, $i = 1\ldots n$ denotes the labels for x_i. The weights are obtained by minimizing $L(w)$, given in (6).

$$L(w) = \frac{1}{2}||w||^2 + C\sum_{1}^{N}\xi_i \tag{6}$$

Subject to contraints:

$$y_i(w^T\phi(x_i) + b) \geq 1 - \xi,\ and\ \xi \geq 0, i = 1\ldots n \tag{7}$$

Here, the value of the bias is b, and the mapping from the input vector to the feature vector is done by $\phi(x)$. It is a problem of maximization of the equation given in (8).

$$Q(\lambda) = \sum_{1}^{n} \lambda_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} y_i y_j \lambda_i \lambda_j \kappa(x_i, x_j) \tag{8}$$

With conditions $\Sigma_i^n y_i \lambda_j = 0$ and $0 \geq \lambda_i \leq C$, $i = 1 \ldots n$. Here C is known for the regularization parameter, which negotiates between the overhead and classification error. The support vectors are the pairs of x_i, which are located at either side of the hyper-planes, and they also define the decision functions. The kernel function transforms the input feature space into higher dimensional space. Here RBF kernel function is used because the classes are mainly spherical. The RBF function is given as: $\kappa(x_i, x_j)$ $=\phi(x_i)\dot{\phi}(x_j)$$= e^{-\gamma|x_i - x_j|^2}$, where γ is weight. The flow diagram of the proposed Multi-Class SVM is given in Fig. 4. It can be seen from the diagram that, input image x is going through five different classifiers having RBFs as kernel functions and produce binary outputs $f_i(x)$, where $i \in [1, 5]$. Five outputs are then fused in $argmax_i\{f_i(x)\}$ and produces name of the disease class.

3 Results and discussions

All the experiments are performed on the platform with the following specification:

- Processor:Intel Core i5, 2.30GHz
- RAM: 8GB

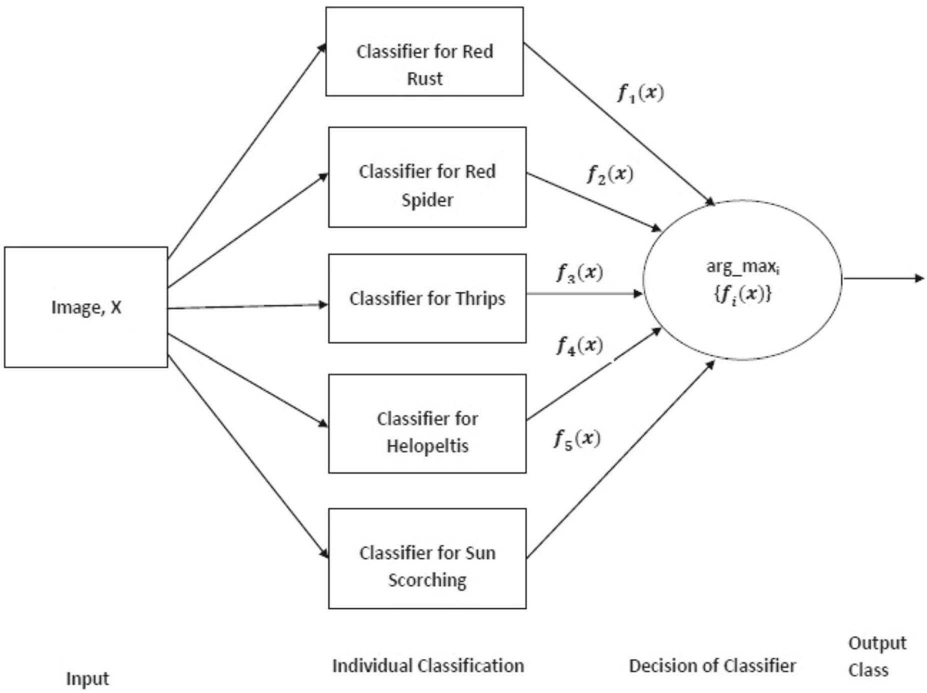


Fig. 4 Flow diagram of the proposed multi-class SVM classifier

- Operating System: 64-bit Windows 10
- Coding Environment: Pyhton 3.7.3

A database is developed from three different tea gardens of the Cachar district, Assam, India, namely Silcoorie, Haticherra, and Rose Candy, by using a Mobile with an 8MP camera. Out of total 312 image samples of five different diseases, 144 Red Rust images, 27 number of Red Spider, 45 number of Helopeltis, 49 number of Thrips, and 47 number of Sun Scorching images are obtained. The acquired images are converted into the Hue Saturation band. The HS band image is segmented using NSGA-II based image clustering algorithm. In the NSGA-II algorithm, the following parameters are used:

- Number of chromosomes: 20
- Number of genes: K
- Number of generations: 100
- Tournament size: 2
- Mutation probability: 0.05
- Crossover probability: 0.9

The NSGA-II gives optimal results when the population size is taken as 20 or more, and the total number of iteration is more than 70 ([48]). This research considers the binary tournament strategy to efficiently solve the problem by avoiding premature convergence ([16]). The size(K) of each chromosome is a user parameter. K represents the number of output clusters. It is known that the probability of mutation should be a very low value, whereas the probability of crossover should be a very large value.

The segmentation algorithm is validated by Silhouette Index (SI) [29]. It takes values between -1 to 1. The higher value of SI illustrates better cluster separation. The output of image conversion from RGB to HS format, segmented images, and Silhouette images of five different images is shown in Fig. 5. The first row contains the outputs on Red Rust Image. It can be seen from the images that the HS image gives the true and saturated colors in the image, which is useful here. From the segmented image, it can be noticed that other portions are infected other than the light blue color, which can be distinguished from this output. The silhouette plot validates the NSGA-II based clustering algorithm. Here it can be seen that very few points are negative or misclassified. Similarly, in the second row, Red Spider image is shown. In this case, only the yellow portions are mainly unhealthy. In the subsequent rows, Helopeltis, Sun Scorching, and Thrips diseased images are shown. In all these images, it can be seen that the segmented images can properly distinguish the unhealthy regions from the leaf images.

After that, the features are extracted from the segmentation outputs, which are Number of Cluster Centers, Inter-Cluster Distance, Variance, Entropy, and Skewness, respectively. The first two features are taken from the chromosomes' fitness values of the proposed NSGA-II based clustering algorithm. Those two feature values are calculated in the previous stage of this methodology. This minimizes the run time complexity of the feature selection algorithm. The correlation matrix is shown in Table 1. It can be noticed from the table that there is a little correlation among the features, except variance and Inter-class distance values. This feature set can provide an accuracy of up to 92% in the case of Red Rust.

The K-fold accuracy score is shown in Fig. 6. It can be noticed there that the average accuracy is obtained more than 83% using the SVM classifier, using K=8.

The under fitting and over fitting validation test is shown in Fig. 7. If the training and the validation scores are both low, the system will be under-fitting. If the training score is high and the validation score is low, then the system is over-fitting, and otherwise, it is

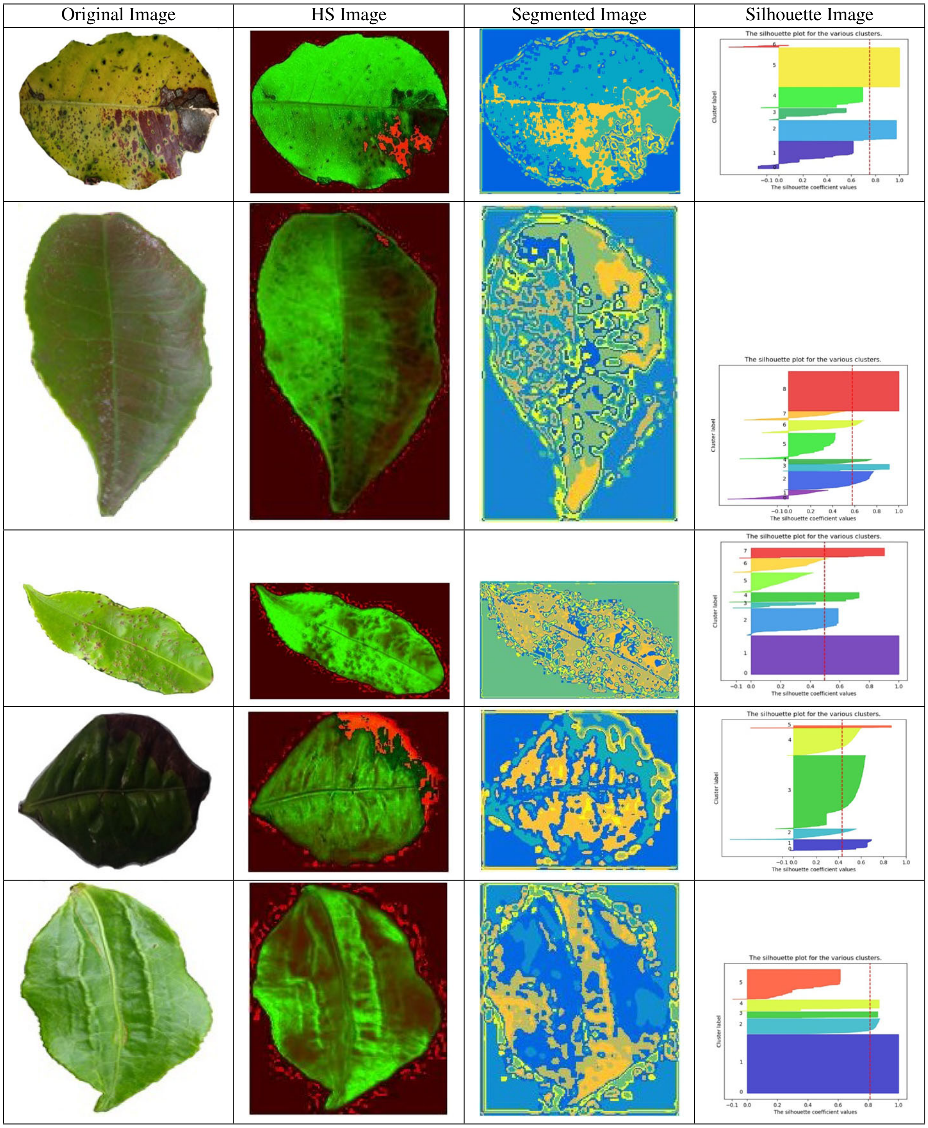


Fig. 5 Output images of all subsequent phases of the algorithm

Table 1 Correlation values among features of tea leaves

Correlation	Variance	Entropy	Skewness	Inter-cluster distance	NoC
Variance	1.000000	-0.263329	-0.017704	0.366119	0.059704
Entropy	-0.263329	1.000000	0.101280	0.159259	0.204124
Skewness	-0.017704	0.101280	1.000000	0.060943	0.002657
Inter-cluster distance	0.366119	0.159259	0.060943	1.000000	0.056181
NoC	0.059704	0.204120	0.002657	0.056181	1.000000

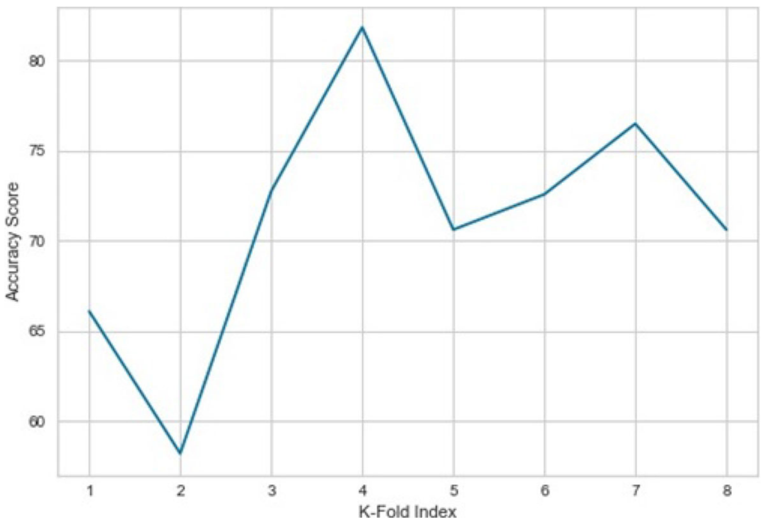


Fig. 6 k-Fold Accuracy Test

working well. In our case, it can be mentioned that the proposed model is working pretty well because the training score and the testing score are not much high or low; instead, they are a bit similar. So, it can be said that the proposed algorithm is free from under-fit and over-fit problems to some extent.

The Precision, Recall, and F1-Score values for the proposed SVM based classifier are computed here. These parameters are prevalent to validate the classification algorithms. The Precision tells about the accuracy of positive prediction, whereas the Recall is the fraction of

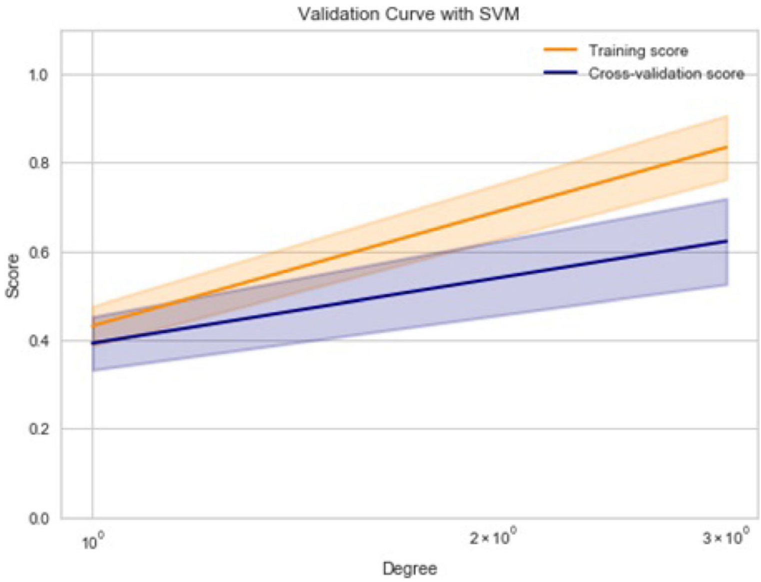


Fig. 7 Underfit/ Overfit Validation Curve

the correctly identified positives. The F1-Score takes into account Precision and the Recall, both. It is obtained by finding the harmonic mean of Precision and Recall. The accuracy is calculated as the number of all correct predictions divided by the size of the data. In this proposed model, the Precision, Recall, F1-Score, and average accuracy values are obtained as 77%, 84%, 78%, and 83% respectively, for the five different leaf images.

The comparisons of the accuracy of the proposed algorithm with K-Means based disease detection algorithm is shown in Fig. 8. The comparisons are performed concerning all individual diseases. It can be seen from the figure that for Red Rust disease, the proposed algorithm is obtaining over 92% accuracy, whereas K-Means based algorithm is 83%. For Helopeltis disease, the proposed algorithm is outperforming K-Means by a wide margin. For Thrips and Sun Scorching, NSGA-II based algorithm obtained over 81% accuracy and which is much better than K-Means based clustering algorithm. But for Red Spider only, the proposed method is not performing well. Here the accuracies are 75% and 73%, respectively. The reason for such a result is the insufficient size of data samples.

In Table 2, experimental results are studied and compared with fifteen existing algorithms. The contributions are written in columns, and in individual rows, the comparisons among the fifteen algorithms are shown. NSGA-II based image segmentation algorithm is considered here for identifying the diseased spots. Five diseases at a time are considered here, which are actually predominant in the study area. The numbers of features used for training the model are identified very carefully so that the training and disease identification become computationally less. The symptoms by which disease identification is made here are of early stage, or even before any mature symptoms become visible. The labeled database generated here is also useful for other researchers, in case if required. The execution time is also an important parameter when a service is offered to stakeholders. The proposed algorithm runs much quicker than existing algorithms. In this table, tick and cross marks are given against eight contribution heads and compared with fifteen existing models, which shows the effectiveness of the proposed model. The proposed model outperforms others in all aspects of the contributions being considered.

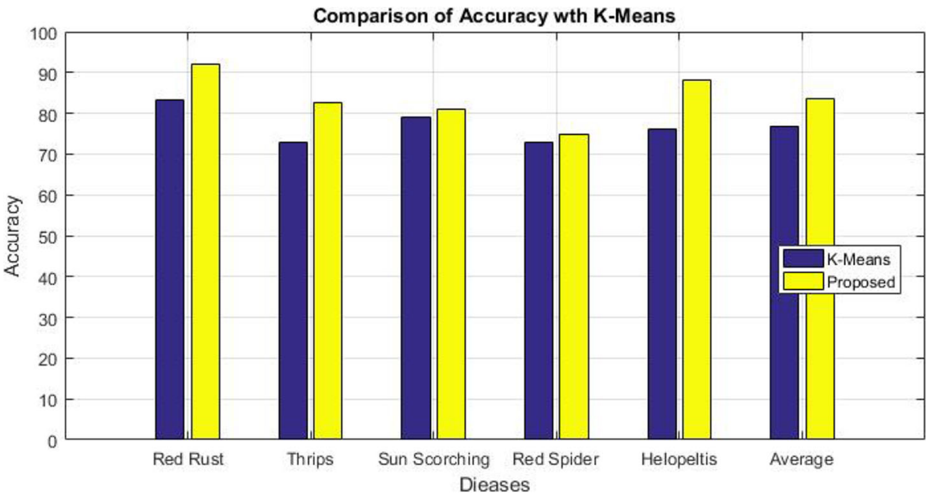


Fig. 8 Comparison of accuracy with K-Means algorithm

Table 2 Tick/Cross comparisons with existing algorithms

Sl. No.	Contributions against Models	Minimum Image Processing	NSGA-II based Segmentation	Reduced number of features	Early stage detection	Real Dataset	Low execution overhead	Web-based "In-Field" service	Tea leaf-based detection
1	Arivazhagan et al. [2]	✗	✗	✓	✓	✗	✗	✗	✗
2	Hossain et al. [18]	✗	✗	✗	✗	✓	✗	✗	✗
3	Karmokar et al. [22]	✗	✗	✗	✗	✗	✗	✗	✓
4	Omrani et al. [42]	✗	✗	✗	✓	✗	✗	✗	✗
5	Padol et al. [43]	✗	✗	✗	✗	✗	✗	✗	✗
6	Phadikar et al. [46]	✗	✗	✗	✗	✗	✗	✗	✗
7	Rumpf et al. [47]	✗	✗	✗	✓	✗	✗	✗	✗
8	X Sun et al. [56]	✗	✗	✓	✗	✗	✗	✗	✗
9	Amar et al. [9]	✗	✗	✗	✓	✓	✓	✗	✓
10	Singh et al. [54]	✗	✗	✗	✗	✓	✗	✗	✗
11	Shanwen et al. [62]	✗	✗	✗	✓	✓	✗	✗	✗
12	Jing et al. [5]	✗	✗	✓	✗	✓	✗	✗	✓
13	Xiaoxiao et al. [55]	✗	✗	✗	✗	✓	✗	✗	✓
14	Shrivastava et al. [53]	✗	✗	✗	✗	✓	✗	✗	✗
15	Zhu et al. [63]	✗	✗	✗	✓	✓	✗	✗	✗
16	**Proposed**	✓	✓	✓	✓	✓	✓	✓	✓

4 Conclusion

In this paper, we have proposed a tea leaf disease detection model, which is based on computationally intelligent algorithms such as NSGA-II, PCA, and multi-class SVM. A real database is developed in the study area. Unhealthy tea leaf images are acquired with early-stage symptoms to restrict the further spread of diseases. The proposed NSGA-II based image clustering algorithm is validated by the Silhouette index. After that, an optimal set of features is chosen using PCA, and multi-class SVM based disease detection is done. The proposed model can detect five different diseases in tea leaves. K-Fold validation, Tick/Cross comparisons, under-fit/over-fit validation, Correlation matrix, and comparisons of accuracy with K-Means, are performed to validate the proposed algorithm. Here an online system is developed to be used "in-field" by the farmers and other stakeholders.

In the future, by generating an extensive database with an equal number of diseased leaves, better accuracy can be achieved. The clustering algorithm with improved fitness functions will be attempted in the future for better segmentation results. In the future, Hyperspectral plant images will be considered for remotely detecting diseases in tea leaves.

Acknowledgments Authors express a deep sense of gratitude towards the authorities of Tea Gardens such as Rose Candy, Haticherra, and Silcoorie, Cachar, Assam, where the data set was obtained, which is the backbone of this work. Without their sincere supports, it was never possible to work in this area.

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